Chlorine

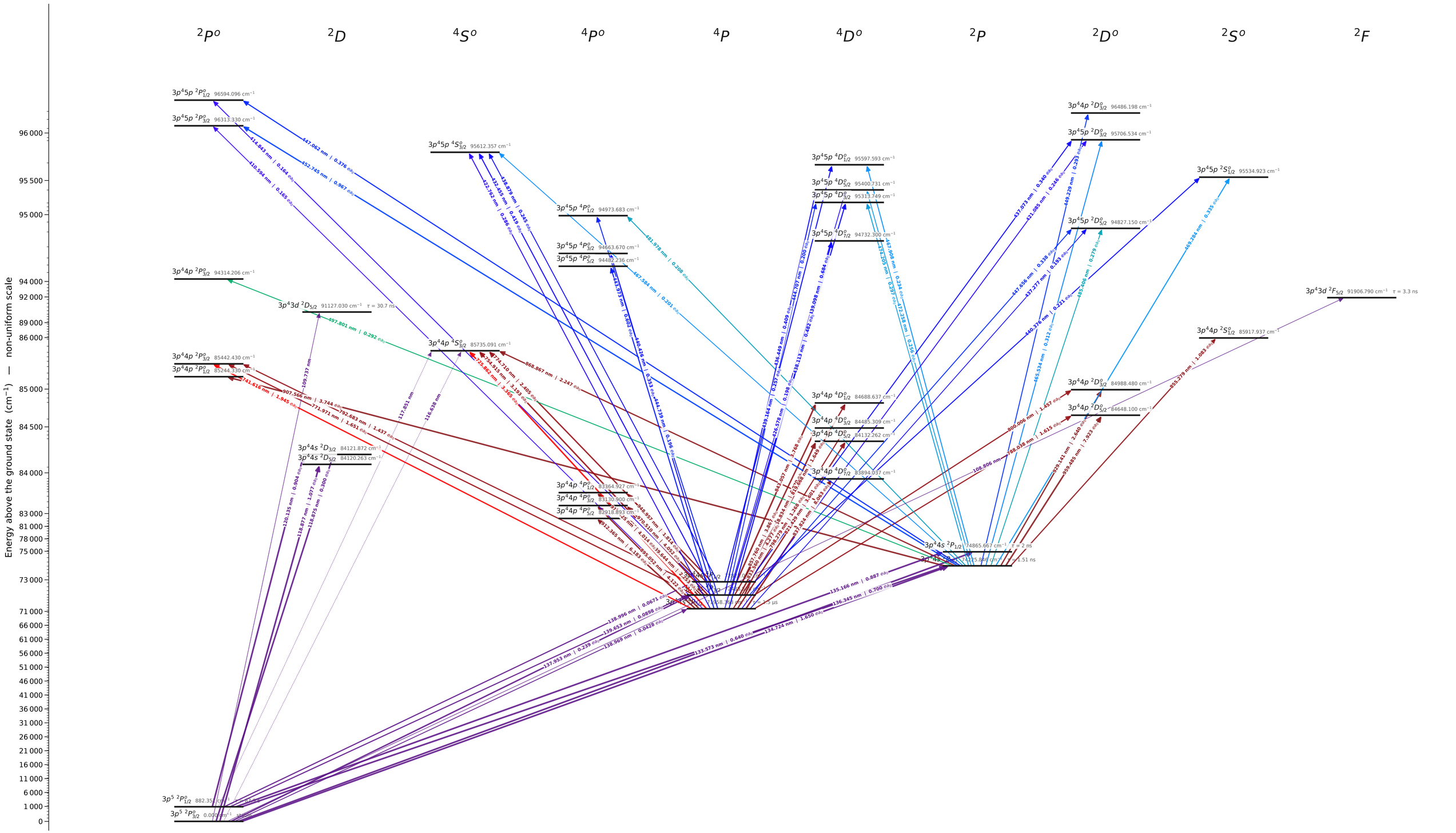
Cl (Cl I). Strongest classified transitions below 2 μm (NIST): 39 levels, 74 lines.

Cl⁺ ionisation limit 104591.01 cm^{-1} = 12.96763 eV (95.611 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\|\langle J\|er\|J'\rangle\|$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
$3p^5\ ^2P^o_{3/2} - 3p^4 3d\ ^2F_{5/2}$	108.8059	–	2755.2962	–	–	48.23	–	interstellar Cl I 1088 A line
$3p^5\ ^2P^o_{3/2} - 3p^4 3d\ ^2D_{5/2}$	109.7369	–	2731.9196	–	–	5.184	–	interstellar Cl I 1097 A line
$3p^5\ ^2P^o_{3/2} - 3p^4 4p\ ^4S^o_{3/2}$	116.6384	–	2570.2734	–	–	–	–	two-photon LIF
$3p^5\ ^2P^o_{1/2} - 3p^4 4p\ ^4S^o_{3/2}$	117.8512	–	2543.8211	–	–	–	–	TALIF detection of spin-orbit excited Cl
$3p^5\ ^2P^o_{3/2} - 3p^4 4s\ ^2P_{3/2}$	134.7240	–	2225.2349	1.650	5.640e+08	105.4	85.2	134.72 nm VUV resonance line
$3p^5\ ^2P^o_{1/2} - 3p^4 4s\ ^2P_{3/2}$	136.3447	–	2198.7827	0.700	9.780e+07	105.4	14.8	136.34 nm VUV line from the metastable 2P1/2 level

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{35}Cl hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$3p^5\ ^2P^o_{3/2}$	205.04687(3)	54.87290(6)	F=0: -700.34 F=1: -550.16 F=2: -194.94 F=3: +475.07
$3p^5\ ^2P^o_{1/2}$	1037.209(14)	–	F=1: -1296.51 F=2: +777.91

Measured A, B constants; sources in the data file.